04 R包安装3

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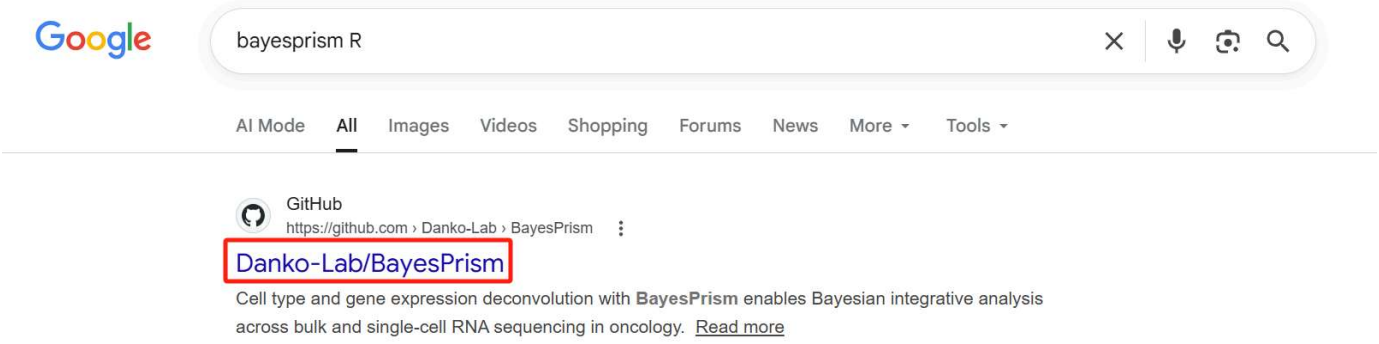
写在前面：暂无：)

安装RTools

- RTools是一个用来从源文件（也就是代码文件）编译生成`R`包的工具链，有许多包的安装都依赖这个工具（比如上节课安装失败的`bluster`），所以我们要提前安装一个。它的下载地址是：https://cran.rstudio.com/bin/windows/Rtools/rtools45/rtools.html。
  - 我已经在本地准备好了一个安装包（05 R包安装4文件夹下的rtools45-6768-6492.exe）

安装方式3

- 如果在CRAN和bioconductor里都找不到想要安装的包，可能就要去github上去找了：



  - 他举的例子在github上的主页为：https://github.com/Danko-Lab/BayesPrism
- 这种包的安装方式如下：

> ### 4 Installation
>
> - R packages:
>
> snowfall, NMF, gplots, scran, BiocParallel, Matrix
>
> - If all dependent packages and commands have been installed, please use the following codes to install/update the package in R terminal.
>
> ```
> library("devtools");
> install_github("Danko-Lab/BayesPrism/BayesPrism")
> ```

  - 具体的命令是：

```
library("devtools");
install_github("Danko-Lab/BayesPrism/BayesPrism")
```

- 当然，首先你要先安装`devtools`这个包：

```
install.packages("devtools")
```

- 下载的时候很可能遇到这个问题，这是因为github需要翻墙访问。解决方法可以参考06 R包安装4 课程（将github文件下载到本地再安装），也可自己翻墙下载：

```
✓ Updated metadata database: 6.03 MB in 15 files.
✓ Updating metadata database ... done

→ Will install 1 package.
→ Will download 1 CRAN package (434.82 kB).
+ remotes   2.5.0 [dl] (434.82 kB)
i Getting 1 pkg (434.82 kB)
✓ Got remotes 2.5.0 (i386+x86_64-w64-mingw32) (436.46 kB)
✓ Downloaded 1 package (436.46 kB) in 2.3s
✓ Installed remotes 2.5.0  (140ms)
✓ 1 pkg: added 1, dld 1 (436.46 kB) [51.4s]
Using GitHub PAT from the git credential store.
Downloading GitHub repo Danko-Lab/BayesPrism@HEAD

Error in .rs.downloadFile(url = url, destfile = path, method = method,  :
  download from 'https://api.github.com/repos/Danko-Lab/BayesPrism/tarball/HEAD' failed
此外:

警告信息:
`install_github()` was deprecated in devtools 2.5.0.
i Please use pak::pak("user/repo") instead.
This warning is displayed once per session.
Call lifecycle::last_lifecycle_warnings() to see where this warning was generated.
```

- 安装本文件夹下的翻墙软件（efan-1.6.2.exe），然后登陆我的账号，账号密码是文件夹下的proxy_account.txt文件。
- 挂上梯子后，记住你把梯子开到了哪个端口：

- 可以这样设置代理：

```
# 设置本地代理（需要先挂梯子）
## 设置HTTP代理
Sys.setenv(http_proxy = "http://127.0.0.1:7897")
## 设置HTTPS代理
Sys.setenv(https_proxy = "http://127.0.0.1:7897")
```

- 然后就可以按照上述命令运行安装啦：

```
Warning: replacing previous import 'BiocParallel::register' by 'NMF::register' when loading 'BayesPr
ism'
** help
*** installing help indices
** building package indices
** testing if installed package can be loaded from temporary location
Warning: replacing previous import 'gplots::lowess' by 'stats::lowess' when loading 'BayesPrism'
Warning: replacing previous import 'BiocParallel::register' by 'NMF::register' when loading 'BayesPr
ism'
** testing if installed package can be loaded from final location
Warning: replacing previous import 'gplots::lowess' by 'stats::lowess' when loading 'BayesPrism'
Warning: replacing previous import 'BiocParallel::register' by 'NMF::register' when loading 'BayesPr
ism'
** testing if installed package keeps a record of temporary installation path
* DONE (BayesPrism)
> library(BayesPrism)

载入需要的程序包：snowfall
载入需要的程序包：snow
载入需要的程序包：NMF
载入需要的程序包：registry
载入需要的程序包：rngtools
载入需要的程序包：cluster
NMF - BioConductor layer [OK] | Shared memory capabilities [NO: windows] | Cores 2/2

载入程序包：‘NMF’

The following object is masked from ‘package:generics’:

    fit
```

- 总结：前三节课讲的都是**代码安装**的方式，通过调用`R`里特定的安装指令来安装`R`包，后面会介绍手动安装的方式。